

MOTH — Race 6



Lago di Garda, Malcesine · 4 Jul 2026 · Start 15:55 (local)

Race course — all boats



Fleet

CRO 5087 *

AUS 5250 *

GER 5248

ESP 56

USA 4846

Race winner

- CRO 5087 * finished first in this race.
- 2nd: AUS 5250 * 3rd: GER 5248
- Afternoon breeze; W-L-W-L course.
- Wind, marks and course axis estimated from GPS (no wind sensor in logs).

Instrument note (*)

- CRO 5087, AUS 5250: speed sensor mounted rotated. Heel & heel-stability are rebuilt from the boat's gravity + course — an accurate estimate, not exact to the degree.

Positions at the marks — gains (green) / losses (red)

Finish	Boat	M1	M2	M3	Fin	Δ M1→M2	Δ M2→M3	Δ M3→Fin	Net
1	CRO 5087 *	1	1	1	1	+0	+0	+0	+0
2	AUS 5250 *	2	2	2	2	+0	+0	+0	+0
3	GER 5248	4	4	3	3	+0	+1	+0	+1
4	ESP 56	3	3	4	4	+0	-1	+0	-1
5	USA 4846	5	5	5	5	+0	+0	+0	+0

M1 = windward 1 · M2 = leeward 1 · M3 = windward 2 · Fin = finish. Positive Δ (green) = places gained on that leg.



Fin	Boat	Line pos	Bias loss (m)	DTL gun	DTL -10s	SOG gun	SOG -10s	TWA gun	SOG 1'	Heel stab 1'	VMG 1'
1	CRO 5087 *	2-1	18	5	11	22.1	19.9	77	19.2	6.9	11.1
2	AUS 5250 *	2-3	11	11	28	21.0	17.0	74	19.8	10.3	12.6
3	GER 5248	1-3	20	10	36	18.6	15.7	79	16.9	3.4	9.3
4	ESP 56	2-2	14	17	44	19.2	17.6	67	18.1	4.4	10.7
5	USA 4846	2-3	10	14	36	18.8	17.5	66	18.1	6.5	9.6

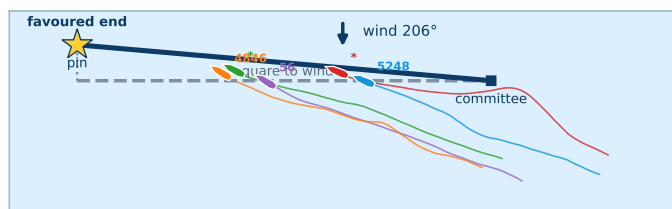
How to read the start

- Line pos: the line is split into 3 thirds, each third into 3 — read third-subpart from the committee end (1-1) to the pin end (3-3).
- Bias loss (m): metres given away at the gun from line position alone vs the favoured (pin) end, from the TWD bias.
- DTL = distance to line (m, negative = over early). 1' = first minute after the gun; heel stability = avg heel change per 10s (lower = steadier).

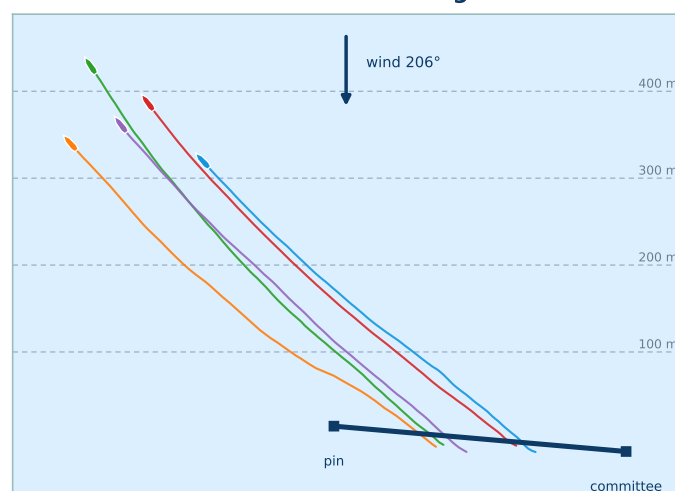
Line bias at the start (vs TWD)

- Start line 338 m, bearing 301°.
- Wind at start ≈ 206°, estimated from the first beat.
- Line bias vs TWD: 5.0° — Pin end favoured.
- Advantage ≈ 29 m (≈ 5.8 boat-lengths) for a boat starting at the pin end.

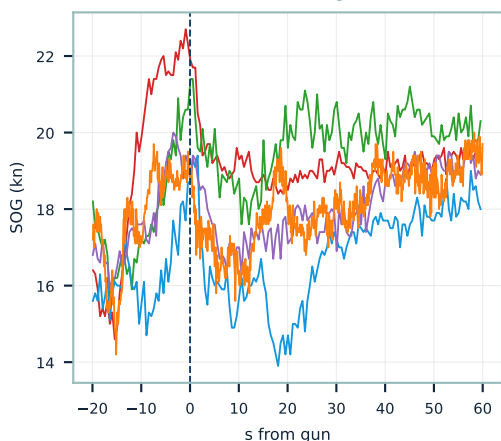
Position at gun



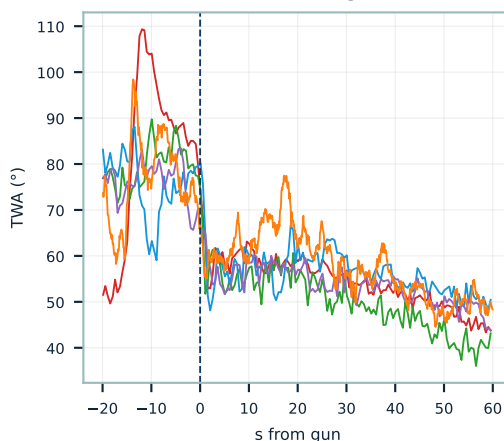
Position 1 min after gun



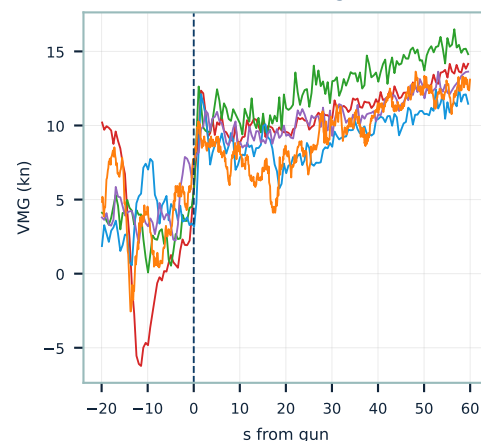
SOG around gun



TWA around gun

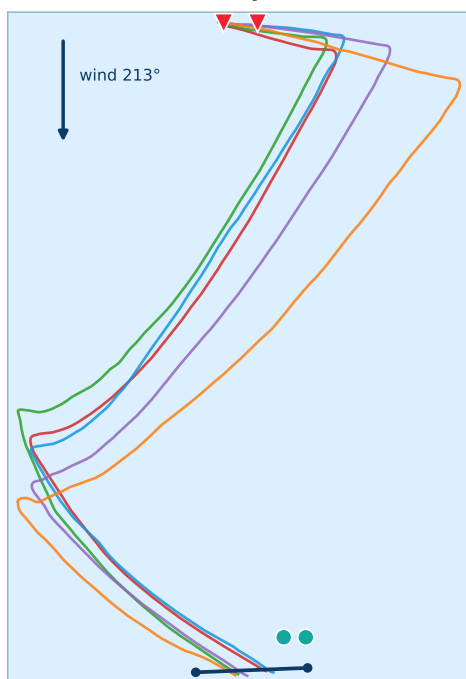


VMG around gun

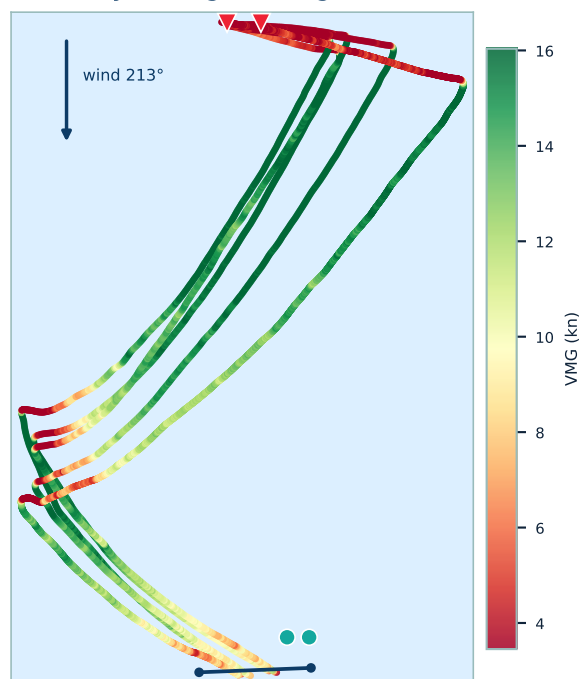




Tracks by boat



Tracks by VMG (green=high, red=low)



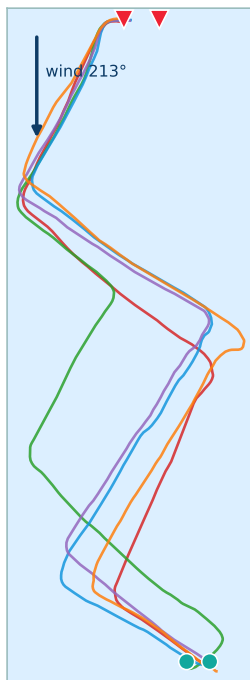
— CRO 5087* — AUS 5250* — GER 5248 — ESP 56 — USA 4846

Finish	Boat	Avg VMG	Avg TWA	Avg SOG	Max SOG	Heel_norm	Heel_stab	Heel Δ	Avg Pitch	Dist (m)	Time (s)	# Tacks
1	CRO 5087*	12.68	46	18.98	20.30	22.2	8.9	-0.0	+0.1	2903	298	2
2	AUS 5250*	12.11	48	17.94	20.94	24.0	9.7	-1.6	-0.5	2901	315	1
4	ESP 56	11.80	49	18.72	20.50	25.5	6.1	+3.9	-0.2	3110	323	2
3	GER 5248	11.32	46	17.17	20.50	26.7	10.3	+2.5	-0.0	2958	335	2
5	USA 4846	9.87	55	17.60	20.20	20.7	7.0	-1.3	-0.5	3507	387	2

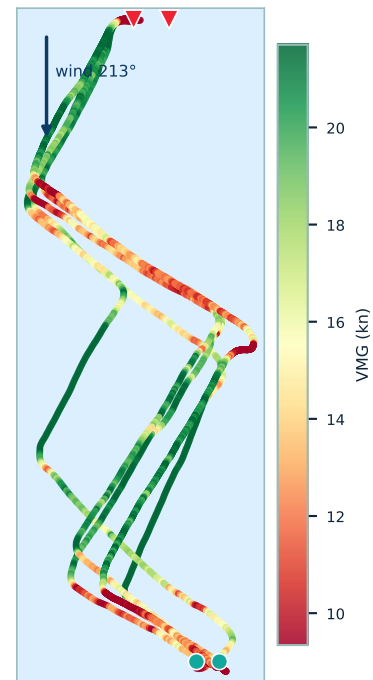
Rows ordered by Avg VMG (best first); the Finish column keeps each boats race finish position. Heel_norm = mean |heel|; Heel_stab = avg heel change per 10s (lower = steadier); Heel Δ = + more heel to starboard / - more to port; Avg Pitch + = bow up; Dist = distance sailed; Time = leg duration.



Tracks by boat



Tracks by VMG (green=high, red=low)



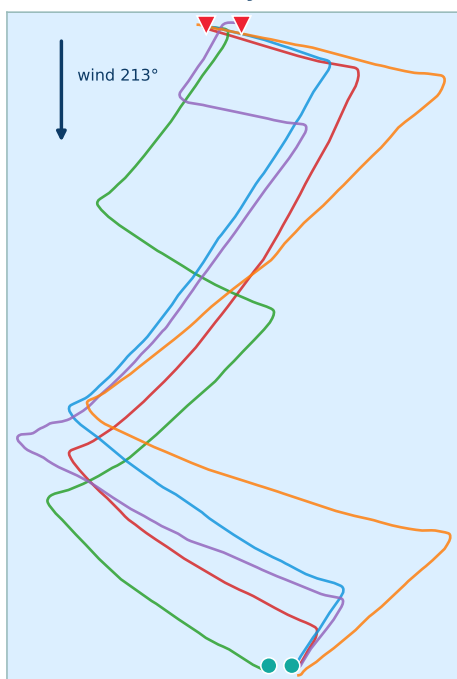
— CRO 5087* — AUS 5250* — GER 5248 — ESP 56 — USA 4846

Finish	Boat	Avg VMG	Avg TWA	Avg SOG	Max SOG	Heel_norm	Heel_stab	Heel Δ	Avg Pitch	Dist (m)	Time (s)	# Gybes
1	CRO 5087*	18.41	143	24.09	26.60	15.5	8.3	+1.8	+0.3	2415	195	3
2	AUS 5250*	18.14	141	24.21	27.40	16.9	12.8	-0.7	+0.3	2481	199	4
3	GER 5248	16.44	139	23.14	26.00	16.2	9.0	+0.0	-0.4	2609	219	3
4	ESP 56	16.00	137	22.62	25.00	13.1	6.9	+0.6	-0.9	2613	225	3
5	USA 4846	14.65	136	20.57	25.00	11.6	6.9	+3.8	+0.6	2643	249	3

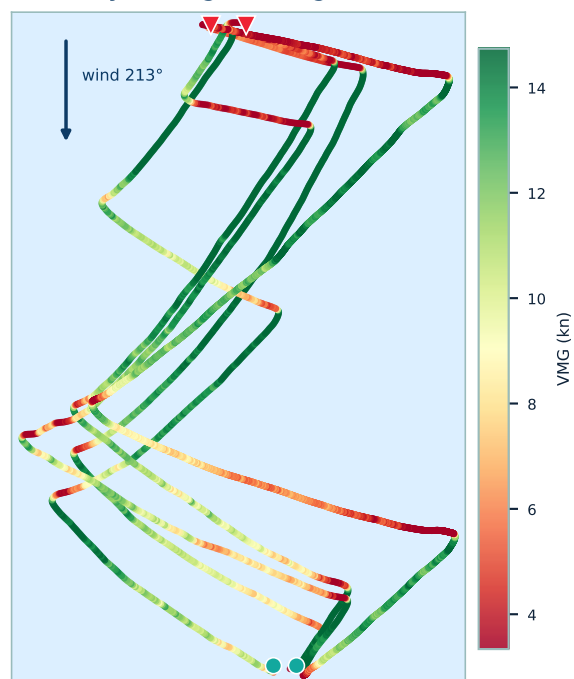
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Tracks by boat



Tracks by VMG (green=high, red=low)



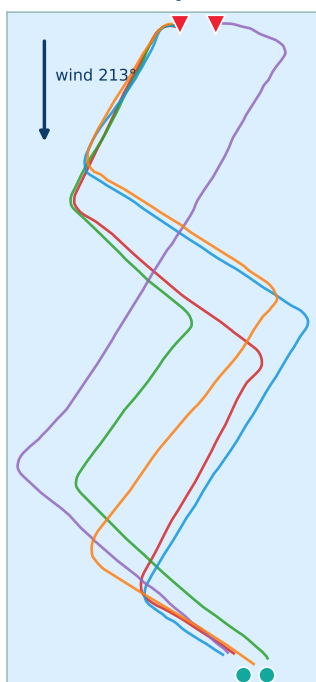
— CRO 5087* — AUS 5250* — GER 5248 — ESP 56 — USA 4846

Finish	Boat	Avg VMG	Avg TWA	Avg SOG	Max SOG	Heel_norm	Heel_stab	Heel Δ	Avg Pitch	Dist (m)	Time (s)	# Tacks
2	AUS 5250*	11.89	50	19.08	21.38	24.5	13.6	-0.7	-0.4	2988	304	4
1	CRO 5087*	11.48	49	18.22	20.20	22.1	9.2	-0.0	+0.6	2925	312	3
3	GER 5248	11.30	49	17.69	19.80	27.7	11.6	+5.8	-0.2	2897	318	3
4	ESP 56	10.07	52	17.65	21.90	24.6	12.1	-0.8	+0.5	3256	359	4
5	USA 4846	8.47	60	17.53	20.80	21.1	8.1	-0.6	-0.1	3888	431	3

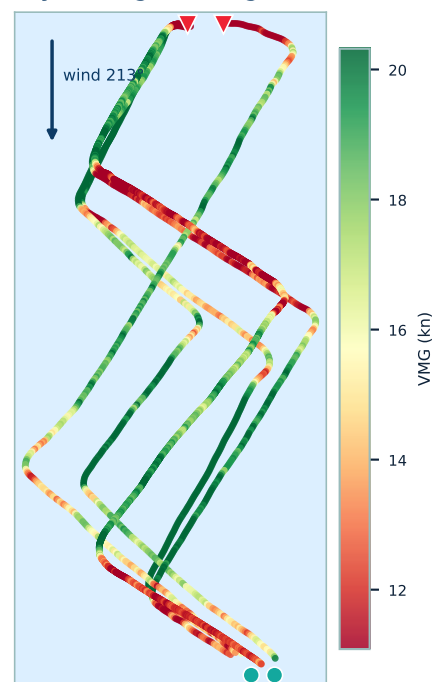
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Tracks by boat



Tracks by VMG (green=high, red=low)



— CRO 5087* — AUS 5250* — GER 5248 — ESP 56 — USA 4846

Finish	Boat	Avg VMG	Avg TWA	Avg SOG	Max SOG	Heel_norm	Heel_stab	Heel Δ	Avg Pitch	Dist (m)	Time (s)	# Gybes
2	AUS 5250*	17.85	139	24.15	26.60	16.2	8.7	-1.2	+0.3	2429	195	3
1	CRO 5087*	17.41	141	23.31	26.30	15.0	6.8	+0.6	+0.2	2381	198	3
4	ESP 56	15.93	139	21.78	24.40	10.7	6.2	-2.0	-0.8	2445	218	2
3	GER 5248	15.75	138	21.99	24.50	16.1	7.2	-0.1	-0.6	2492	220	3
5	USA 4846	15.01	136	21.58	24.50	11.9	8.6	+4.4	+0.5	2617	236	3

Rows ordered by Avg VMG (best first); the Finish column keeps each boats race finish position. Heel_norm = mean |heel|; Heel_stab = avg heel change per 10s (lower = steadier); Heel Δ = + more heel to starboard / - more to port; Avg Pitch + = bow up; Dist = distance sailed; Time = leg duration.



Finish	Boat	# Tacks	Entry	Exit	Min	SOG Δ	Angle Δ	Turn °/s	Heel in	Heel out	Dist loss (m)	VMG loss (m)
2	AUS 5250 *	5	19.1	14.3	12.2	6.9	+22	105	21	22	26.1	45.0
4	ESP 56	6	18.7	13.6	11.6	7.1	+17	109	22	22	27.7	41.1
1	CRO 5087 *	5	18.7	13.5	11.6	7.1	+26	103	20	19	28.2	50.9
3	GER 5248	5	18.2	12.8	10.9	7.3	+26	106	23	22	29.8	49.3
5	USA 4846	5	16.7	9.2	7.3	9.4	+39	54	9	18	41.6	59.8

Rows ordered by Dist loss (most efficient first); the Finish column keeps each boat's race finish position.

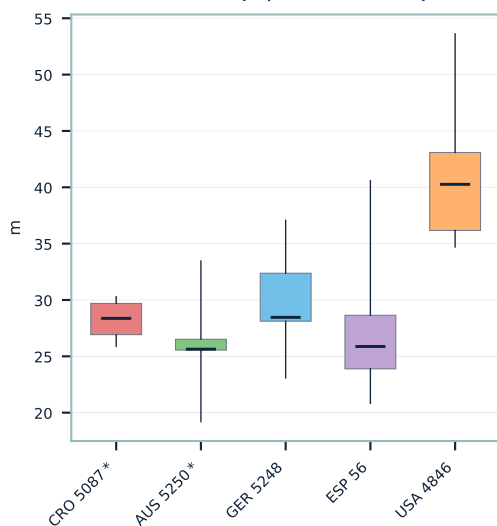
Tacks analysis — metric definitions

- Entry / Exit SOG = mean speed over 6 s before / 9 s after the manoeuvre. Min SOG = lowest speed in between.
- SOG Δ = Entry – Min (how much speed was bled off). Angle Δ = exit TWA – entry TWA (+ = exits wider).
- Rate of turn = peak heading change (°/s). Heel in/out = mean |heel|.
- VMG loss = made-good (toward the mark) deficit vs holding entry VMG. Dist loss = actual distance lost from the speed drop vs holding entry speed.

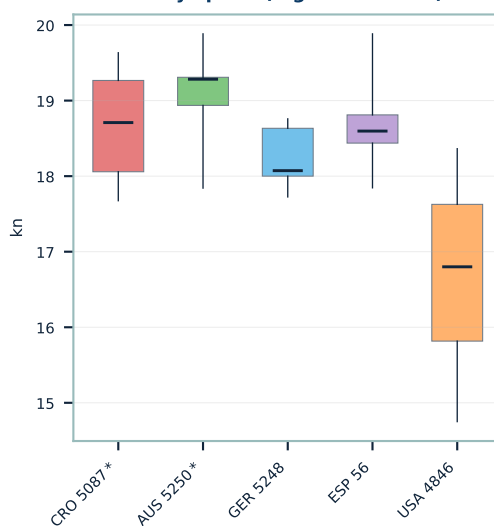
How to read the charts below

- Each box = one boat, ordered left→right by race finish. The plots span all of that boat's manoeuvres.
- Whiskers = min & max; box = middle 50% (25th–75th pct); line in the box = median. A lower, tighter box = more consistent.
- VMG/Dist loss, Speed loss, Min-speed: lower is better. Entry & Min speed: higher is better.

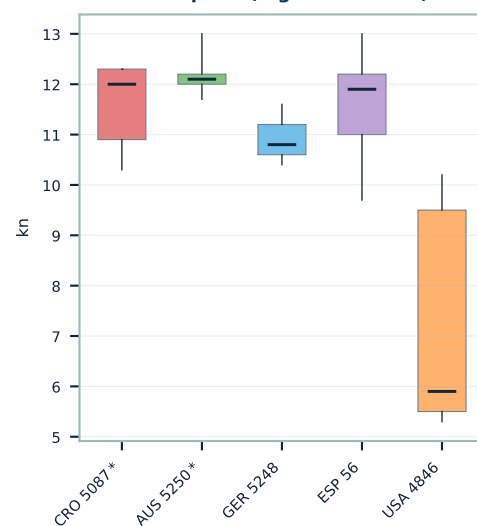
Dist loss (m, lower=better)



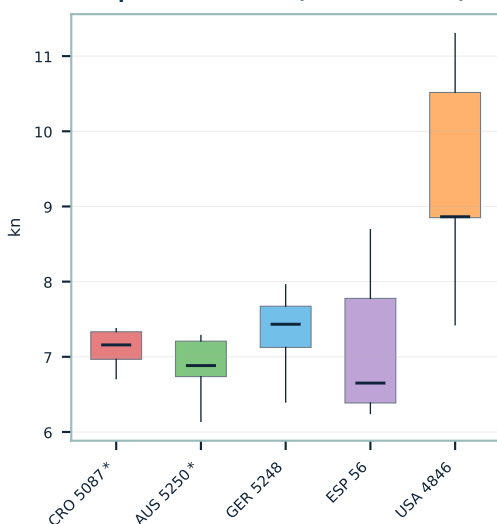
Entry speed (higher = better)



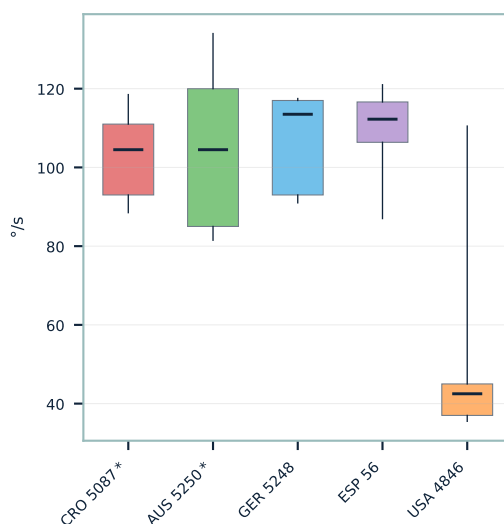
Min speed (higher = better)



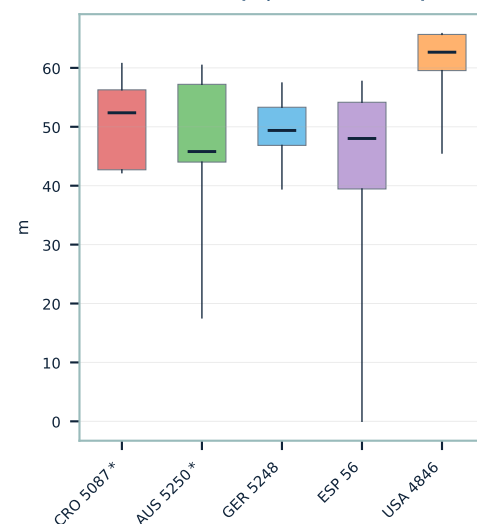
Speed loss SOG Δ (lower = better)



Rate of turn



VMG loss (m, lower=better)





Finish	Boat	# Gybes	Entry	Exit	Min	SOG Δ	Angle Δ	Turn °/s	Heel in	Heel out	Dist loss (m)	VMG loss (m)
4	ESP 56	5	21.5	18.8	16.7	4.9	-1	96	12	9	16.0	15.9
2	AUS 5250 *	7	24.1	21.1	18.8	5.3	-5	77	14	13	17.2	19.6
1	CRO 5087 *	6	23.0	19.1	17.5	5.5	-11	63	11	10	22.1	32.4
3	GER 5248	6	22.1	18.1	16.3	5.8	-6	74	11	10	22.6	28.3
5	USA 4846	6	20.6	14.9	12.6	8.0	-13	40	7	8	32.0	39.5

Rows ordered by Dist loss (most efficient first); the Finish column keeps each boat's race finish position.

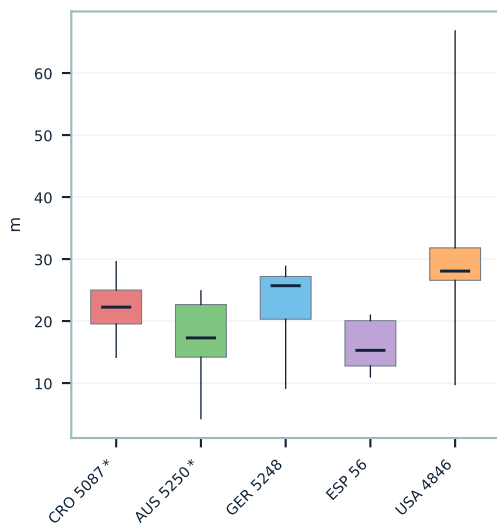
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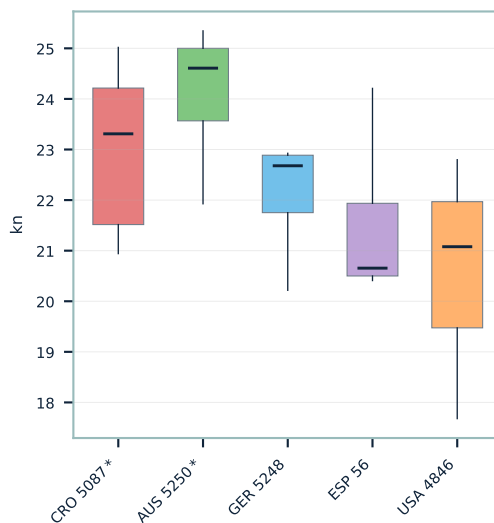
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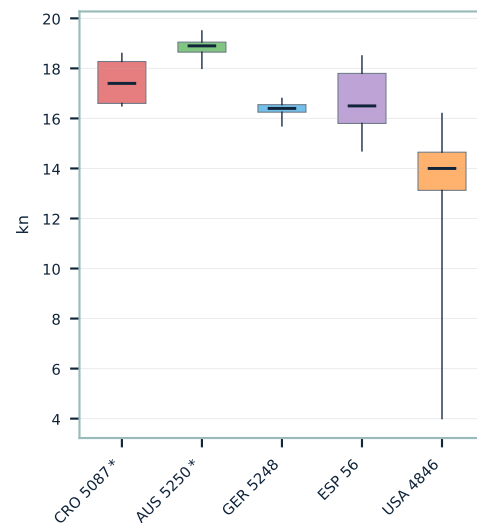
Dist loss (m, lower=better)



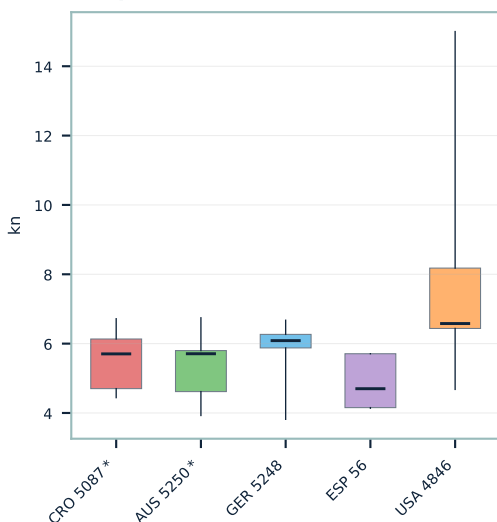
Entry speed (higher = better)



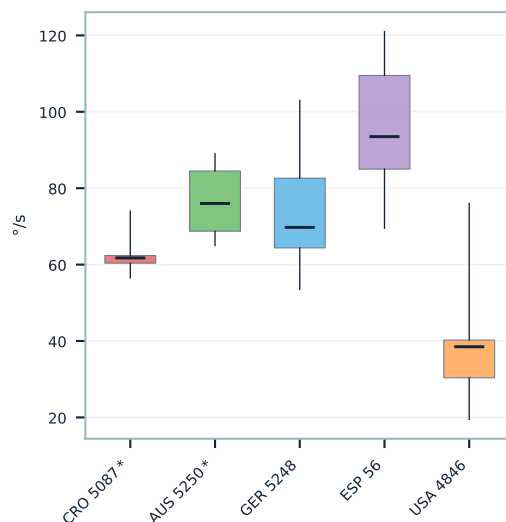
Min speed (higher = better)



Speed loss SOG Δ (lower = better)



Rate of turn



VMG loss (m, lower=better)

